

UML Diagrams & Specifications

1. Use Case Diagram Specifications

- Actor: End-User (Evaluator, Researcher, Administrator)
- Actions (Use Cases):
 1. View Dashboard Analytics
 2. Simulate Scenario (Adjust Sliders)
 3. Run Prediction (Triggers ML Pipeline)
 4. Compare Countries side-by-side
 5. Export PDF Report & Download CSV History
 6. Switch UI Theme (Light/Dark Mode)

2. Class Diagram Specifications

- Database Manager Class: Manages SQLite connections, query insertions, and log history.
- ML Model Handler: Loads Scaler, SVM Classifiers, and Regressors; applies Log(GNI) and Standard Scaling.
- Flask Application App: Sets up routing endpoints (/predict, /compare, /history, /model_stats, /download_report).